

**INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHAGALPUR**

CS204, Database Management Systems,

Spring 2023, Fourth Semester, End-Semester Examination

Time: 8:00 AM to 11:00 AM (180 minutes), Full marks: 50, Date: 25th April, 2023**INSTRUCTIONS:**

- Answer to the questions in order and to the point.
- Assume the missing data suitably and mention the same.

1. a. Explain DBMS overall system architecture.	4+3+3
b. What are the major advantages of database management systems over traditional file system?	
c. What is an entity, instance, schema and Entity-Relationship (ER) model in database systems.	
2. a. Write ER diagram and database schema (with proper Referential Integrity Constrains) for Company Database	5+5
b. Consider the relational database schema as shown below: Give an expression in the relational algebra to express each of the following queries: <i>employee (employee_id, name, date_of_birth, address, salary, dept_No)</i> <i>project (project_No, project_name, project_location, dept_No)</i> i. List the name and salary of all the employees. ii. Retrieve the Name and Salary of all employees who are working for department 1. iii. Find the employees who work for department 3 and whose salary is greater than Rs. 30000. iv. List the name and location of the projects which are not controlled by department 2.	
3. a. Consider the relational database schema as shown below: <i>professor (emp_id, emp_name, start_year, phone_No, dept_No)</i> <i>department (dept_No, dept_name, HoD_empid, phone_No)</i> <i>Courses (Course_id, course_name, credits, dept_No)</i> <i>Slove the following by expressing in terms relational algebra with proper given join operations.</i> i. For each department, find the name and phone number of the head of the department. (Solve by using theta join). ii. Find courses offered by each department. (Solve by using Equi join). b. Differentiate between relational algebra and tuple relational calculus. c. Explain conversion of ER diagrams to Relational Schema for the following cases: case 1: When ER diagram has any weak entity. case 2: When ER diagram has 1:1 binary relationship.	4+2+4
4. a. What is Normalization? Why it is required in database systems?	2.5+5+2.5

b. Explain first normal form (1NF) and Second Normal Form (2NF) with an example.																							
c. Explain Fourth Normal Form (4NF) with an example.																							
5.	a. Explain in detail all the ACID properties.	4+3+3																					
	b. What is serial schedule, non-serial schedule and serializable schedule.																						
	c. Consider the following schedule 'S'. Show whether the given graph is conflict serializable or not by using the precedence graph.																						
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